DAVID ALEXANDER GARCIA MELGAR

SENIOR SOFTWARE ENGINEER | AI ENGINEERING & INTELLIGENT AUTOMATION

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PROFESSIONAL SUMMARY

Senior Software Engineer with 20+ years of experience building enterprise software, API integrations, Python/Node.js automation, and scalable backend systems. Hands-on experience with MCP servers, vector similarity queries for image-data matching, n8n deployment on Linux, and PostgreSQL vector support. Uses Anthropic Claude in IDE workflows, OpenAI Codex, and Gemini in Android Studio for AI-assisted development. Background includes GPS platforms, telecommunications, government integrations, and containerized deployments.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C# (.NET), SQL, PL/SQL, Java, PHP, C++, Kotlin; Ruby (small project exposure)

Frontend & Mobile: Angular, React, Next.js, React Native, HTML, CSS, Android

Backend & AI Integration: Node.js, Express, Python automation, REST APIs, C#/.NET, SOAP, WS-Security, MCP server integrations, vector similarity queries for image data, AI-assisted workflows

Databases & Vector Search: PostgreSQL with vector extension (pgvector) for n8n workflows and similarity queries, Oracle 11g/12c/19c/21c, MySQL, SQL Server, MongoDB, BigQuery; SQL tuning and indexing

AI Development Tools & Infrastructure: Anthropic Claude (IDE), OpenAI Codex, Gemini (Android Studio), n8n (Linux installation/deployment), Docker, Kubernetes, Linux, Nginx, Git, Redis, Kafka, Agile/Scrum

PROFESSIONAL EXPERIENCE

Senior Software Engineer | Sternemedia S. de R.L. | Apr 2014 – Present

- Built production applications, backend integrations, automation workflows, and data-intensive enterprise platforms for telecommunications, GPS fleets, reporting, and government services.
- Worked with engineering teams of up to 20 professionals on application delivery, technical design, complex troubleshooting, code review, and production issues.
- Built and supported operational systems monitoring more than 2,200 telecommunications sites, integrating live operational data, alerts, reporting, and distributed field-team workflows.
- Engineered SQL-intensive applications with Oracle PL/SQL packages, stored procedures, indexes, query optimization, and integrations with Node.js and Angular/React interfaces; also worked with MySQL.
- Implemented cross-border government interoperability using C#, SOAP, and Oracle, reducing integration errors from approximately 48% to 2.5% in a workflow processing about 6,000 exchanges.
- Developed with Python, Node.js, C# and modern web technologies; used Claude, Codex and Gemini for AI-assisted coding; worked with MCP servers and image-data vector similarity queries; configured n8n and PostgreSQL vector support on Linux; deployed and troubleshot Docker/Kubernetes workloads.
- Delivered production solutions used by Tigo (Millicom), Claro, and Digicel; associated supplier evaluations reached 92% in 2023 and 100% in 2024.

PROJECTS

- Telecom GPS & Fleet Monitoring (Eagleview): Developed real-time GPS tracking dashboards, vehicle location and movement reporting, fatigue analytics, alerts, access authorization workflows, and performance-optimized Oracle reporting for distributed field operations.
- Government Digital Services: Built online renewal and international interoperability services, including secure APIs, C#/SOAP interfaces, and Oracle-backed integration with legacy production systems.
- Field Reporting & AI-Assisted Automation: Designed Telegram/web ticketing and evidence-capture workflows with Node.js, Oracle packages, RBAC, configurable templates, and AI-assisted review concepts; distinguished design-stage functionality from deployed capabilities.
- AI Integration, Vector Search & Automation: Worked with MCP servers and vector similarity queries for approximate image-data matching; installed and configured n8n and PostgreSQL vector support on Linux; operated containerized Node.js applications on Kubernetes with database persistence and service networking.

CERTIFICATIONS & EDUCATION

- Oracle SQL & PL/SQL training/certification (2004) | Scrum Master | Microsoft Azure Fundamentals | Cisco CCNA / ICND coursework
- Systems Engineering studies — Honduras 2006
- University of British Columbia —technology 2025